

EESC 214 Earth Science Data Analysis

Autumn Weaver

Fall 2026

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1 Looking at Data

Objectives when first looking at data:

1. Understand the general character of the dataset
2. Understand the general behavior of individual parameters
3. Detect obvious problems with the data

Several steps when analyzing data:

1. Format conversion
Converting format to something easier to analyze. Format conversion is often needed but can introduce error.
2. Reality Checks Check that the data makes sense in context and compared to your expectations.
3. Time Plots Plot the data against time to observe trends.
4. Histograms Plot the distribution of data to observe patterns and potential errors or outliers.
5. Rate Information Plot the rate of change of different parameters.
6. Scatter Plots Plot parameters against each other to observe if there are any correlations.

2 Probability and Measurement Error

Definition 2.1. A **random variable** is a function whose value is determined by a random event.

Data with measurement error is an important case of a random variable.

Definition 2.2. A **probability** $P(d)$ is a function describing the probability that random variable d has a certain value. The total probability of d over its sample space is 1.

Definition 2.3. A **probability density function** $p(d)$ is a function whose integral over a subset of the sample space is the probability that d lies in that section:

$$P(d_1, d_2) = \int_{d_1}^{d_2} p(d) dd$$

Random variables often tend to clump around values. There are three main choices of a typical value: mean, median, and mode.

Definition 2.4. The **mode** of a probability density function $p(d)$ is the point d_{mode} that has the highest probability density.

Definition 2.5. The **median** of a probability density function $p(d)$ is the value d_{median} that divides the sample space in half:

$$\int_{d_{min}}^{d_{median}} p(d) dd = \int_{d_{median}}^{d_{max}} p(d) dd$$

Definition 2.6. The **mean** or **expected value** of a probability density function $p(d)$ is the weighted

average of the p.d.f.:

$$\bar{d} \approx \sum_S^M d^{(S)} P(d_s)$$

$$\bar{d} = \int_{d_{min}}^{d_{max}} dp(d) dd$$

It is also important to measure how tightly the data is clumped around the typical value. This is usually done with variance.

Definition 2.7. The **variance** of a random variable d is:

$$\sigma_d^2 = \int_{d_{min}}^{d_{max}} (d - \bar{d})^2 p(d) dd$$

σ_d is called the **standard deviation** of d .

There are many probability density distributions, but the most important for us are the uniform and normal distributions.

Definition 2.8. The **uniform distribution** has the same probability density everywhere in the sample space:

$$p(d) = \frac{1}{d_{max} - d_{min}}$$

Definition 2.9. The **normal distribution** is:

$$p(d) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(d - \bar{d})^2}{2\sigma^2}\right\}$$

The probability density function of a random variable m that is a function of a random variable d can be determined using the chain rule:

$$P(d_1, d_2) = \int_{d_1}^{d_2} p(d) dd = \int_{d(m_1)}^{d(m_2)} p[d(m)] \frac{\partial d}{\partial m} dm = \int_{m_1}^{m_2} p(m) dm$$

$$\rightarrow p(m) = p[d(m)] \left| \frac{\partial d}{\partial m} \right|$$

The mean and variance of a function of a random variable can be found the same way.

3 Multivariate Distributions